

AoyuX Annual Earthquake Forecasting Report for China Seismic Experimental Site (CSES)

version 1

Issued date:

08:00:00 (AM), January 15, 2025 (Beijing time)

Report type:

Annual report for January 1, 2025 to December 31, 2025 (365 days)

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Report ID:

AoyuX-20250115080000_year_v1

1. Earthquake Forecasts for the Next Year in the CSES

This report uses the earthquake catalog up to and including December 31, 2024, with a forecast time window covering the following year (i.e., from January 1 to December 31, 2025, a total of 365 days). The catalog uses local magnitude (M_L) as the measure of earthquake size. The local magnitudes are published by the China Earthquake Networks Center (CENC). The earthquake forecasts are generated using simulations based on the Epidemic Type Aftershock Sequence (ETAS) model with spatially varying background rates (Nandan et al., 2021a; 2021b, 2022; Li et al., 2024).

It is expected that, in the next year, the level of seismicity within the CSES (Chuan-Yunnan region) for earthquakes of magnitude 3 to 4 will slightly decrease compared to the last year (2024). The level of seismicity for earthquakes of magnitude 5 or larger will remain low with no significant change. The probabilities of at least one earthquake of magnitude 5 or larger, and 6 or larger, occurring in the region are 98% and 43%, respectively.

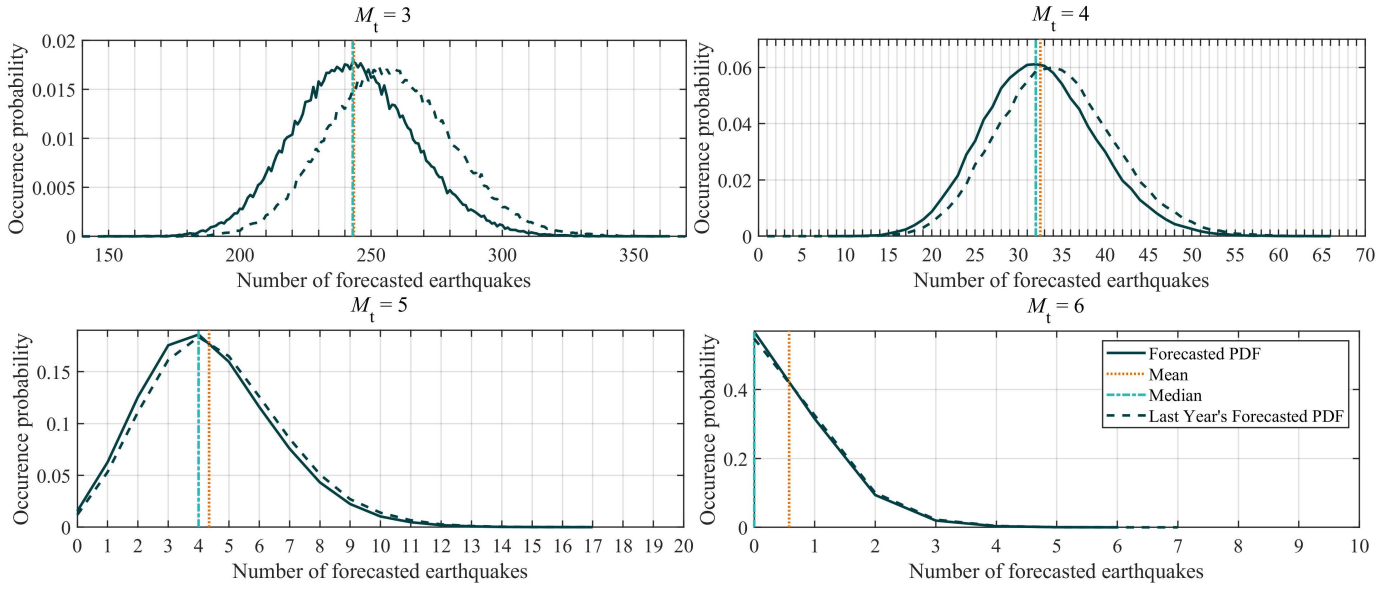


Figure 1. Forecasted probabilities of the numbers of earthquakes of different magnitude ranges ($M \geq M_t$) for the next year.

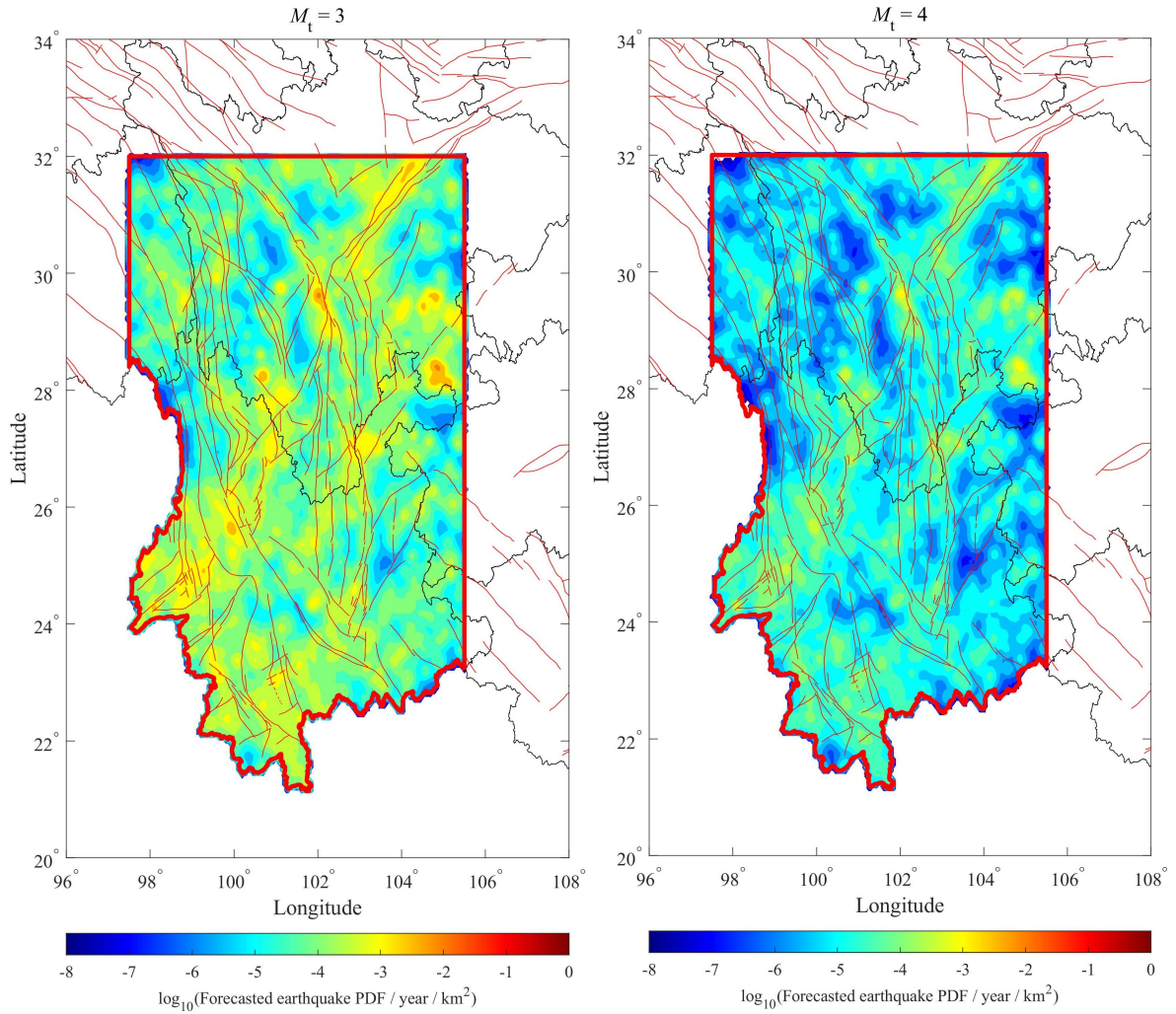


Figure 2. Probability density distribution (PDF) of the numbers of forecasted earthquakes ($M \geq M_t$) for the next year per unit area (km^2) and per year.

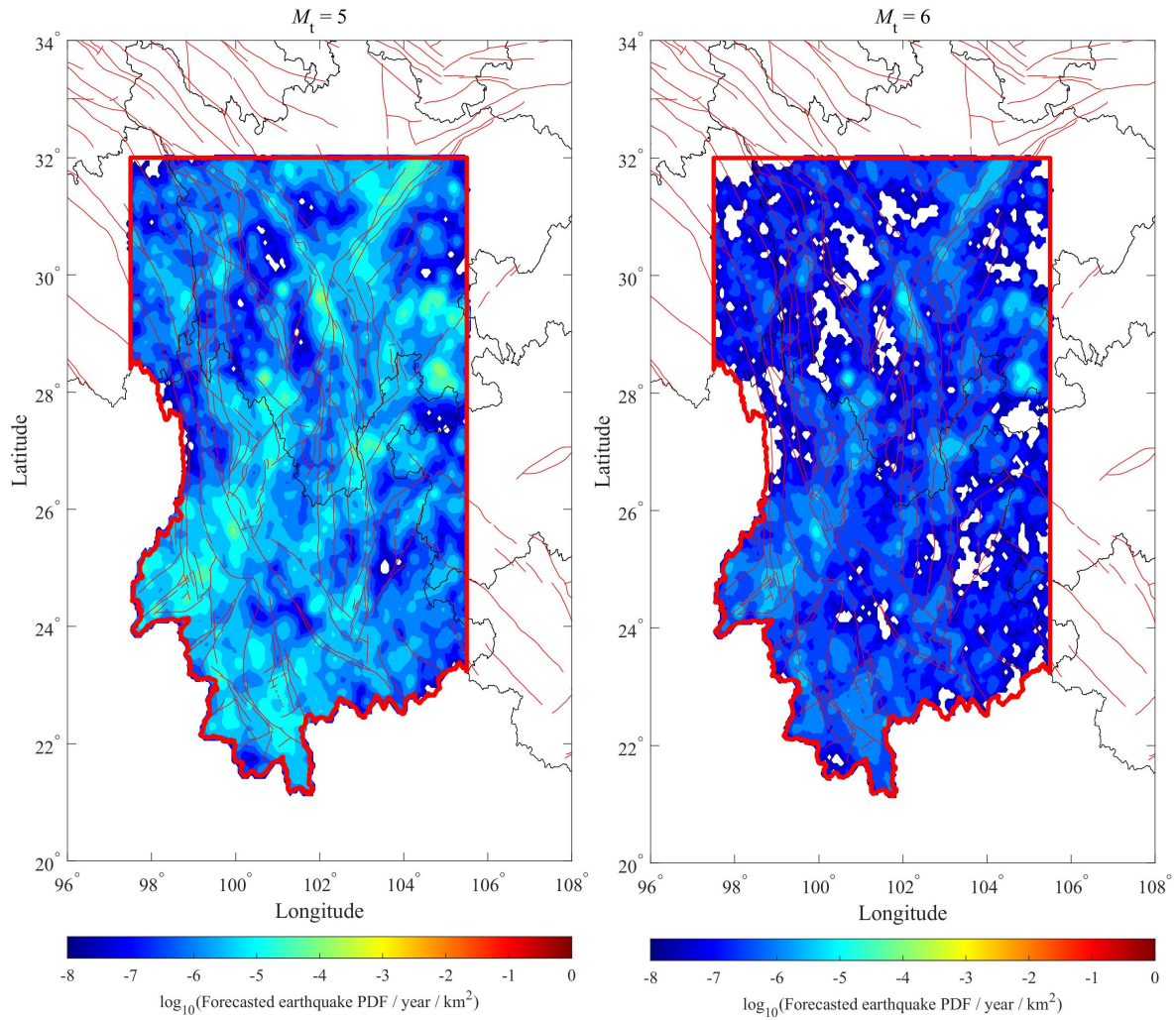


Figure 2. -continued.

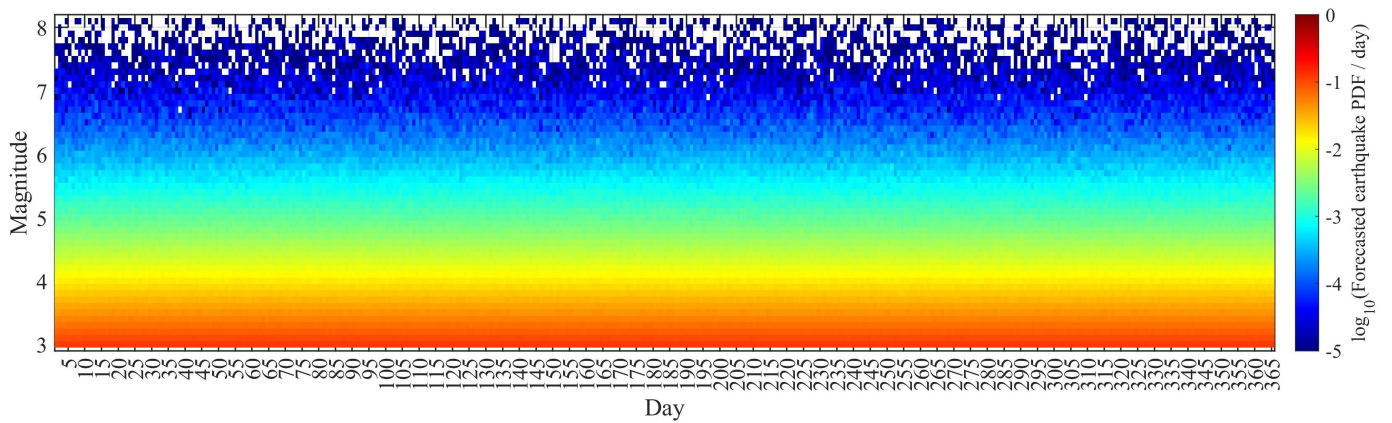


Figure 3. Daily probability density distribution (PDF) of the number of forecasted earthquakes per day over the whole CSES region for the next year.

2. Last Year's Forecast *versus* Realised Seismicity

In the last year (i.e., from January 1 to December 31, 2024, a total of 366 days), the region has experienced 254 earthquakes of $M \geq 3$, 28 earthquakes of $M \geq 4$, 6 earthquakes of $M \geq 5$, with no earthquakes of $M \geq 6$.

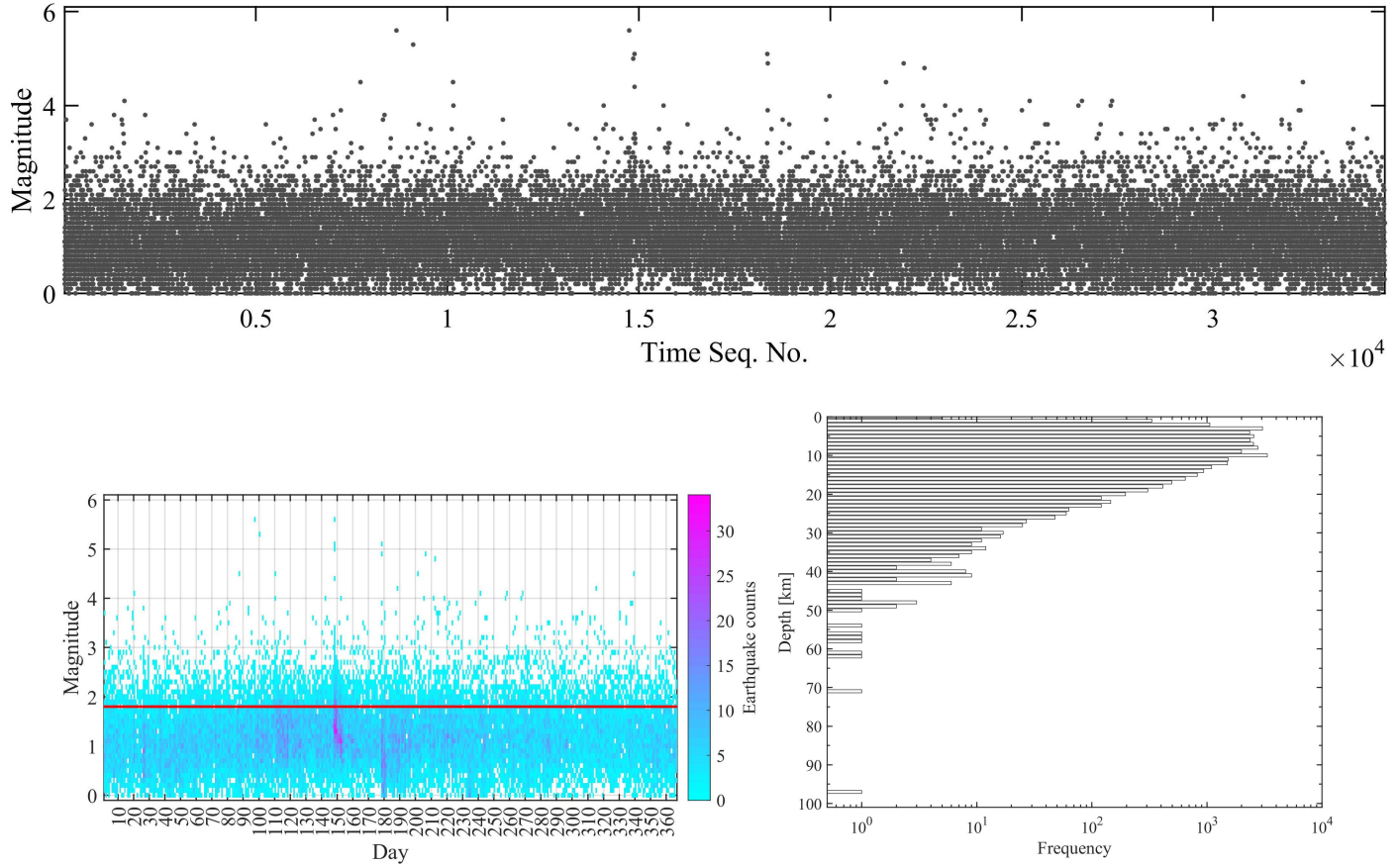


Figure 4. Seismicity in the CSES over the last year. (Top) Magnitude-time sequence plot; (Bottom left) Daily earthquake counts per 0.1 magnitude bin. (Bottom right) Depth-frequency distribution of earthquakes. The red line in the bottom-left figure represents the completeness magnitude (M_c) of the catalog from the last year.

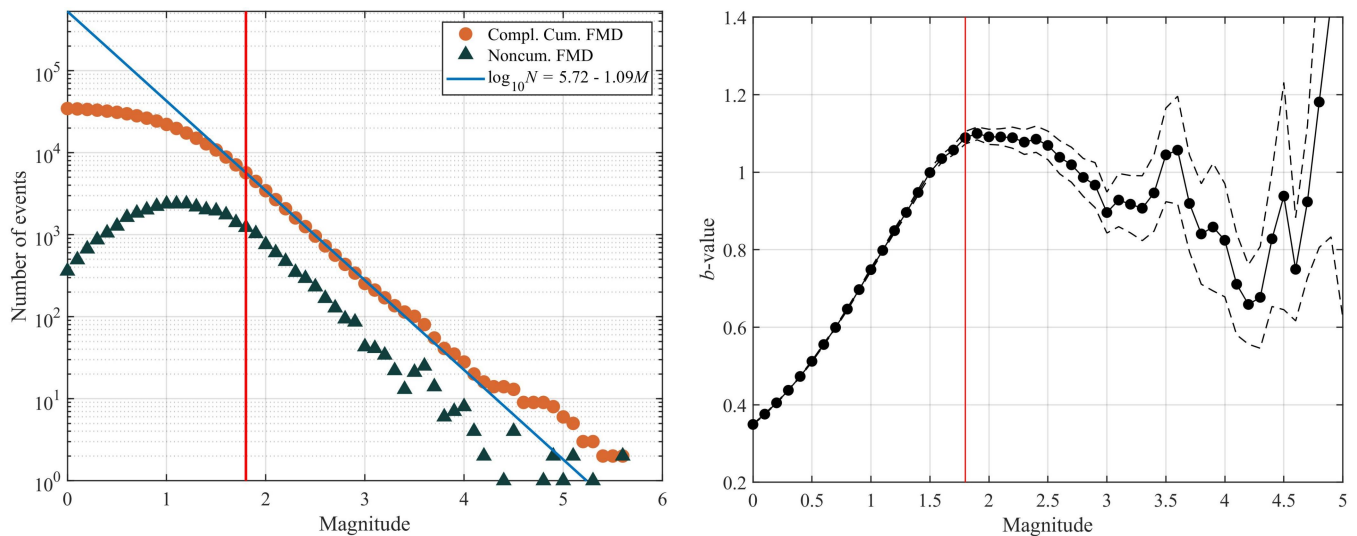


Figure 5. (left) Complementary cumulative and density frequency-magnitude distribution of seismicity from the last year in the CSES, alongside the Gutenberg-Richter (GR) law fitted using earthquakes with $M \geq M_c$, where M_c is derived for the last year (red vertical line); (right) Variation of the b -value of the GR law with M_c , with the thin dashed lines indicating the $\pm 1\sigma$ standard deviation boundaries.

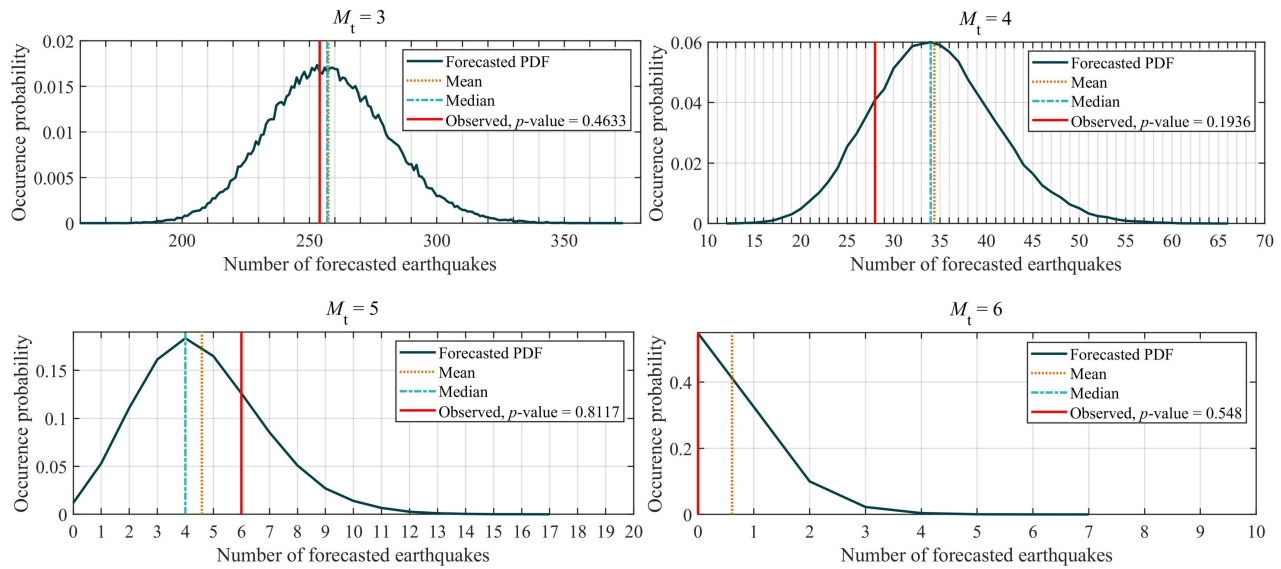
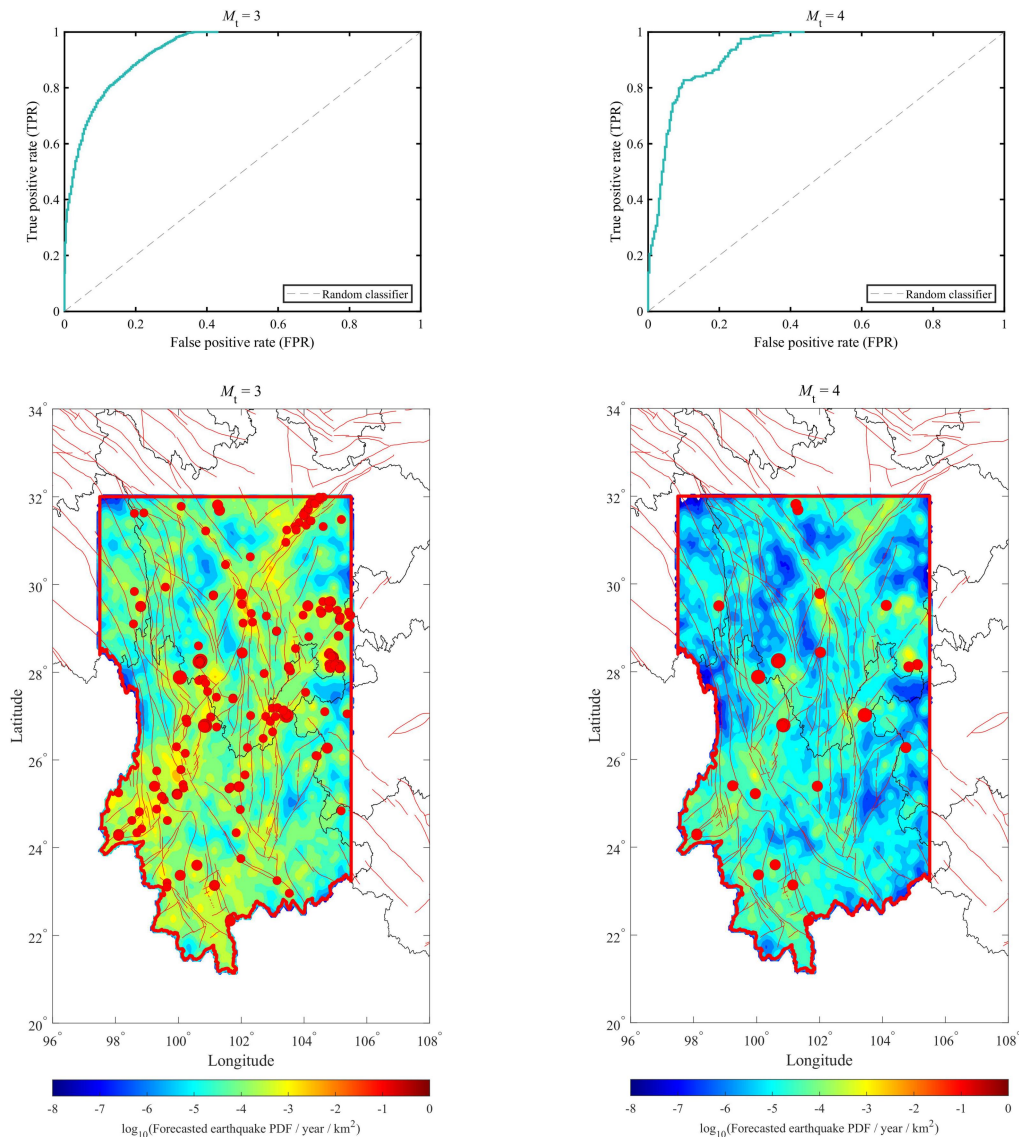


Figure 6. Number of observed earthquakes ($M \geq M_t$) in the CSES over the last year (red vertical line) and forecasted probabilities for earthquakes of different magnitude ranges ($M \geq M_t$) published in the last year. The p -value is the cumulative distribution function (CDF) value for the number of observed earthquakes.



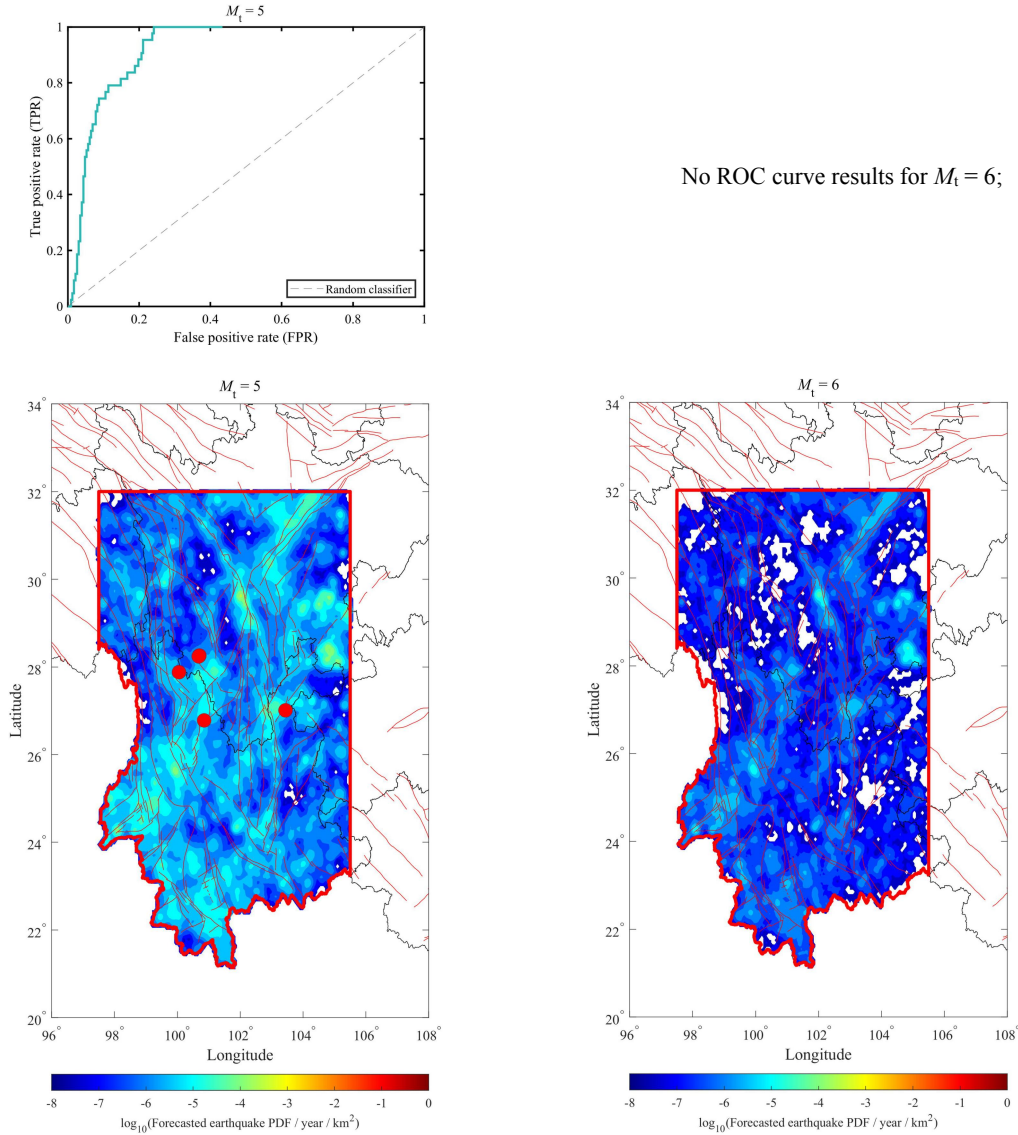


Figure 7. Top: Receiver Operating Characteristic (ROC) curve. The four following panels show the probability density distribution (PDF) of forecasted earthquakes ($M \geq M_t$) for the same period per area (km^2) and per year, released by the last report as the background. Observed seismicity over the last year are shown as red solid circles. The ROC curve is constructed by iterating through all forecasted earthquake probability density values in descending order. For each probability density threshold, the true positive rate (TPR) and false positive rate (FPR) of the grid cells where actual earthquakes occurred are calculated. In this case, the ROC curve is built based on the probability density distribution of forecasted earthquakes ($M \geq M_t$) per area, as published in the last annual forecast report. This process involves comparing the forecasted probability density with the actual earthquakes that occurred during that period to evaluate the performance of the prediction model.

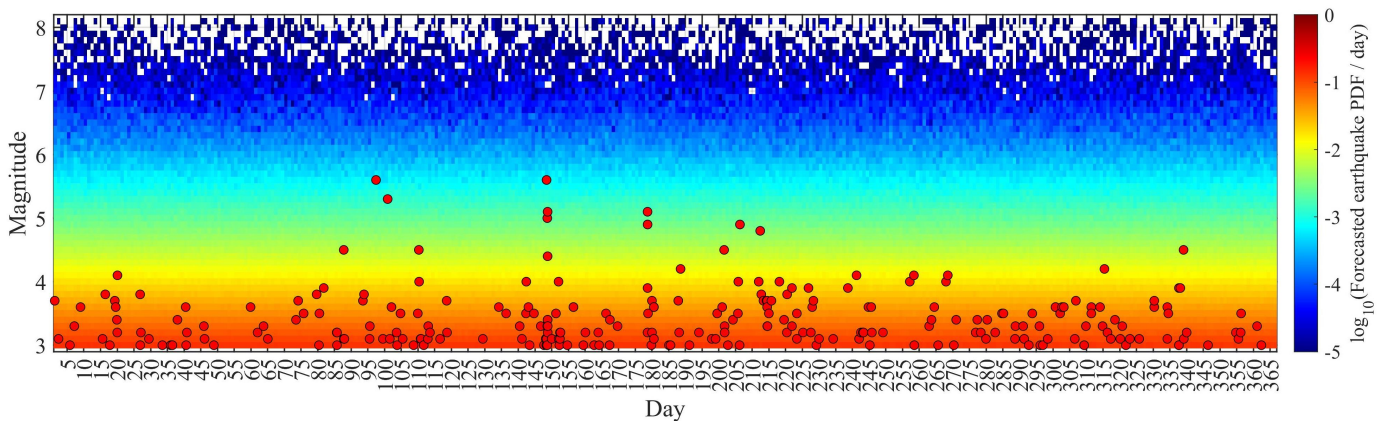


Figure 8. Observed seismicity (red solid circle) over the last year, with the daily probability density distribution (PDF) of the numbers of forecasted earthquakes for the same period, released by last report as the background.

3. Background Information

For more detailed background information on the region, please refer to the AoyuX Monthly Earthquake Forecasting Report.

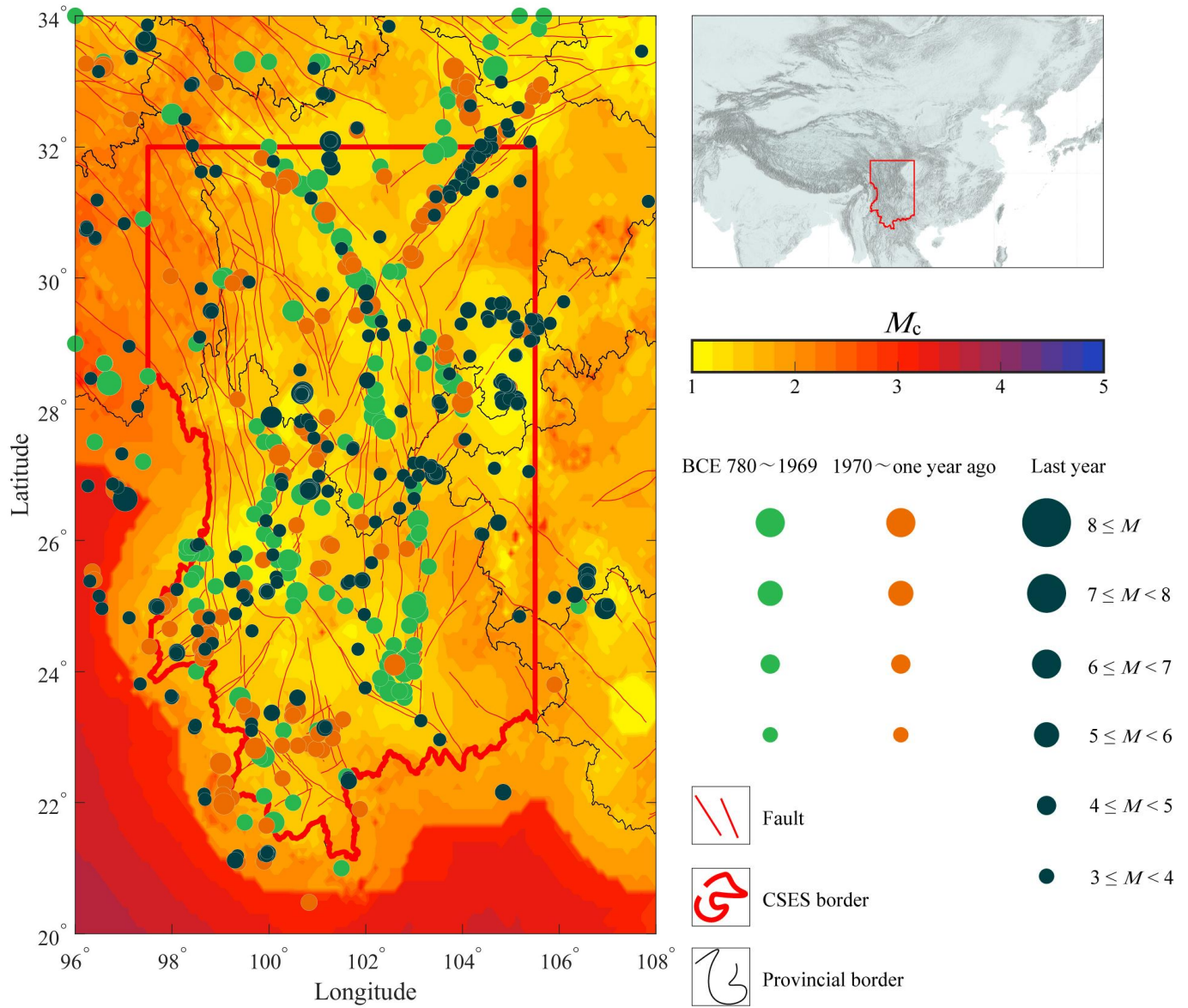


Figure 9. Spatial distribution of earthquakes and faults in the CSES located in the Sichuan-Yunnan region. The background map, showing the completeness magnitude (M_c), is adapted from Li et al. (2024).

Acknowledgement

Since the first release of the AoyuX Earthquake Forecasting Series Report on June 4, 2024, we have received many constructive comments and suggestions from colleagues. These colleagues, in the order their feedback was received, include Prof. Jiancang Zhuang, Prof. Zhongliang Wu, Associate Prof. Yu Feng, Prof. Guixi Yi, Prof. Changsheng Jiang, Prof. Yongxian Zhang and Associate Prof. Ke Wu. We express sincere gratitude to all of them! Most of these comments and suggestions have been reflected in the subsequently released series reports.

The earthquake catalogs in this report are sourced from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at the China Earthquake Administration, available at <http://10.5.160.18/console/index.action> (last accessed: December 18, 2023), with a Digital Object Identifier (DOI) of 10.11998/SeisDmc/SN. The Bayesian Magnitude Completeness (BMC) method used for calculating the spatial distribution of M_c was provided by Dr. Arnaud Mignan. The Epidemic Type Aftershock Sequence (ETAS) model code for simulating earthquake catalogs and calculating earthquake probabilities was derived from a code shared by Dr. Shyam Nandan, which is used in (Nandan et al., 2021a;b; 2022). Fault data were obtained from the GMT Chinese Community at <https://docs.gmt-china.org/latest/dataset-CN/CN-faults/> (last accessed: May 31, 2024). This work is partially supported by the Guangdong Basic and Applied Basic Research Foundation (Grant No. 2024A1515011568), and the Center for Computational Science and Engineering at Southern University of Science and Technology.

References

- Li, J., Mignan, A., Sornette, D., & Feng, Y. (2024), Predicting the Future Performance of the Planned Seismic Network in Chinese Mainland. *Seismological Research Letters*, 94(6), 2698–2711.
- Li, J., Sornette, D., Wu, Z., Zhuang, J., & Jiang, C. (2024). Revisiting Seismicity Criticality: A New Framework for Bias Correction of Statistical Seismology Model Calibrations. (Submitted to *Journal of Geophysical Research* or arXiv preprint arXiv:2404.16374).
- Nandan, S., Kamer, Y., Ouillon, G., Hiemer, S., & Sornette, D. (2021a), Global models for short-term earthquake forecasting and predictive skill assessment. *The European Physical Journal Special Topics*, 230(1), 425–449.
- Nandan, S., Ram, S. K., Ouillon, G., & Sornette, D. (2021b), Is seismicity operating at a critical point? *Physical Review Letters*, 126(12), 128501.
- Nandan, S., Ouillon, G., & Sornette, D. (2022), Are large earthquakes preferentially triggered by other large events? *Journal of Geophysical Research*, 127(8), e2022JB024380.

Attachment: About AoyuX

AoyuX is a developing online research platform dedicated to advancing earthquake predictability in China. It is produced by the Institute of Risk Analysis, Prediction, and Management (RisksX) at the Southern University of Science and Technology (SUSTech), in collaboration with multiple research institutions. In Chinese mythology, "Aoyu" is a legendary creature with a dragon's head and a fish's body, believed by ancient Chinese to cause earthquakes. The "X" represents the unknown and infinite in Western culture, symbolizing goals and hopes.

The AoyuX platform aims to use cutting-edge statistical seismology tools for forward earthquake forecasting research. It will also serve as an online research platform for forward earthquake forecasting competitions with other models. In the future, the AoyuX platform will integrate geophysical observation data, artificial intelligence (AI), and other technologies to conduct various types of earthquake forecasting research, including statistical, numerical, and physical methods. AoyuX is dedicated to help building a modern earthquake forecasting evaluation system in China, encompassing mid-term, short-term and imminent forecasts. It will develop a new visualization platform for earthquake predictability research, facilitating communication between experts and the public, and serving the combined efforts of experts and the public in tracking and assessing the imminent occurrence of major earthquakes, as well as in mid-term and short-term earthquake forecasting.

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